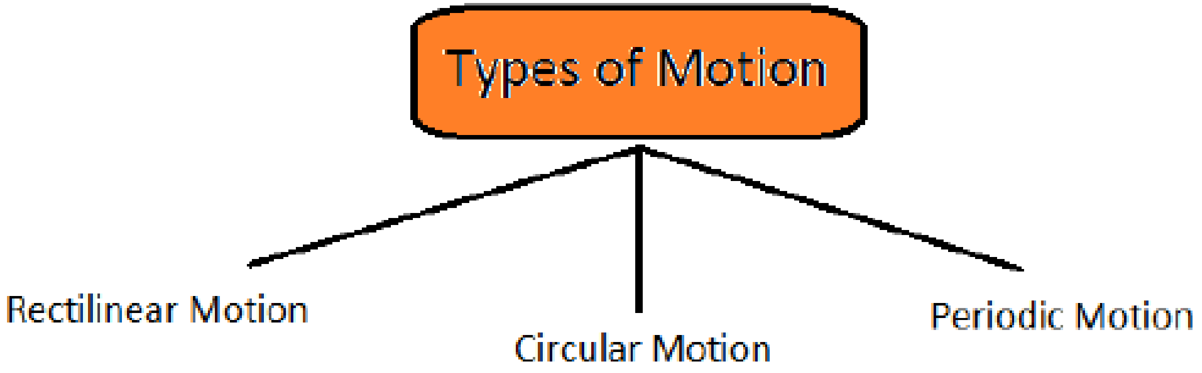
Mechanical actuation systems

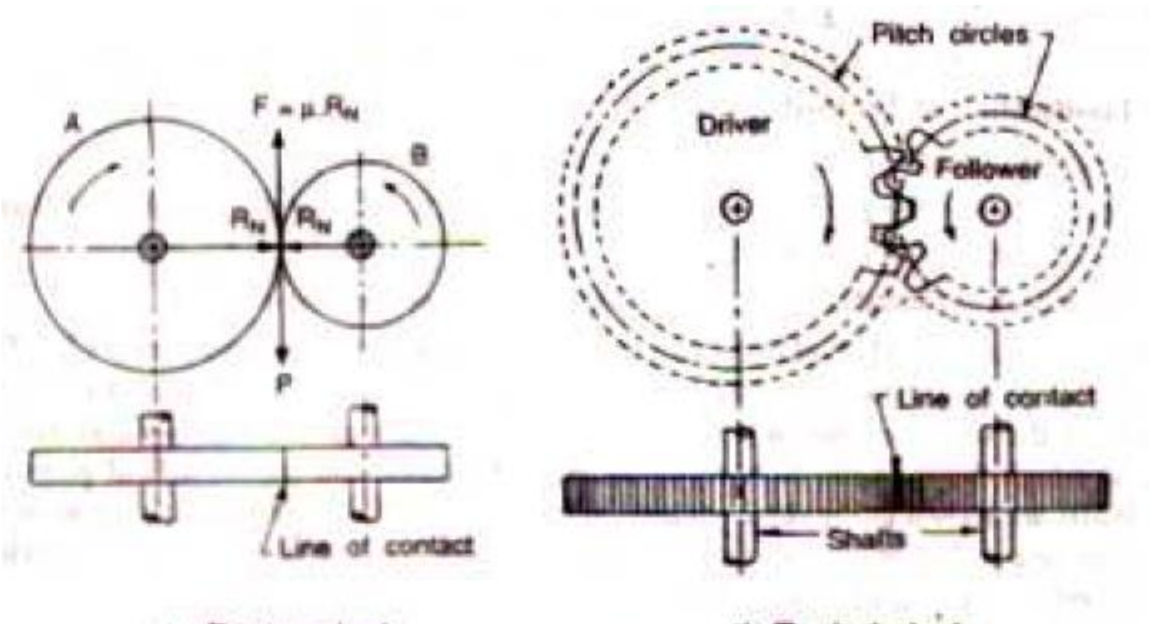
Mechanical systems

- Mechanical Systems are Devices which generate or transform motion from one form to some other required form.
- Transform linear motion into rotational motion and vice versa; A linear reciprocating motion into rotary motion - mechanical elements like linkages, cams, gears, rack-and-pinion, chains, belt drives.
- Rack-and-pinion can be used to convert rotational motion to linear motion
- Types of Motion - The motion of an object shows its changing position



Gears

Gears are machine elements use to transmit rotary motion form one shaft to another with a constant speed ratio.

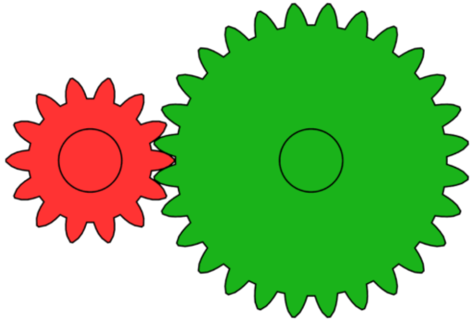


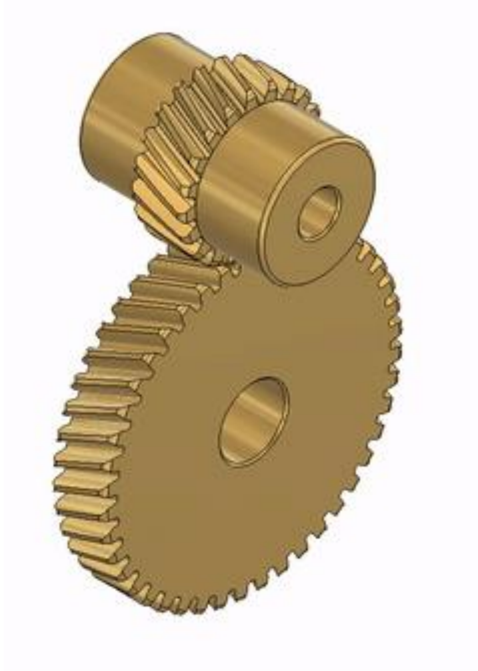
(a) Friction wheels. (b) Toothed wheels.

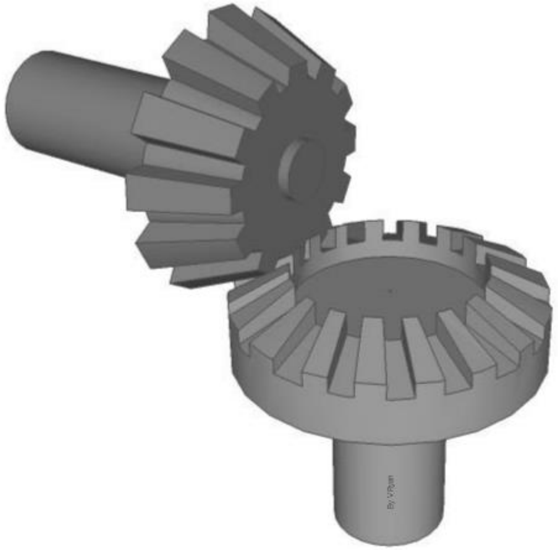
Types of Gears

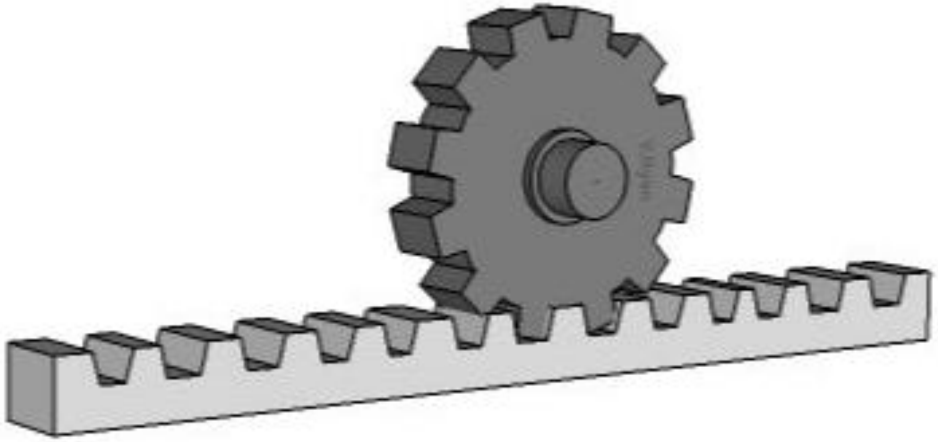
- Spur gears: These are the simplest type of gear. They have straight teeth that are parallel to the axis of the gear
- Helical gears: Helical gears have teeth that are cut at an angle to the axis of the gear.
- Bevel gears: Bevel gears are used to change the direction of rotation between two shafts that intersect at an angle.
- Worm gears: Worm gears are used to convert rotary motion into linear motion, or vice versa.
- Rack and pinion: linear actuator that consists of a straight gear and a pinion gear as the pinion gear rotates, the rack moves linearly.

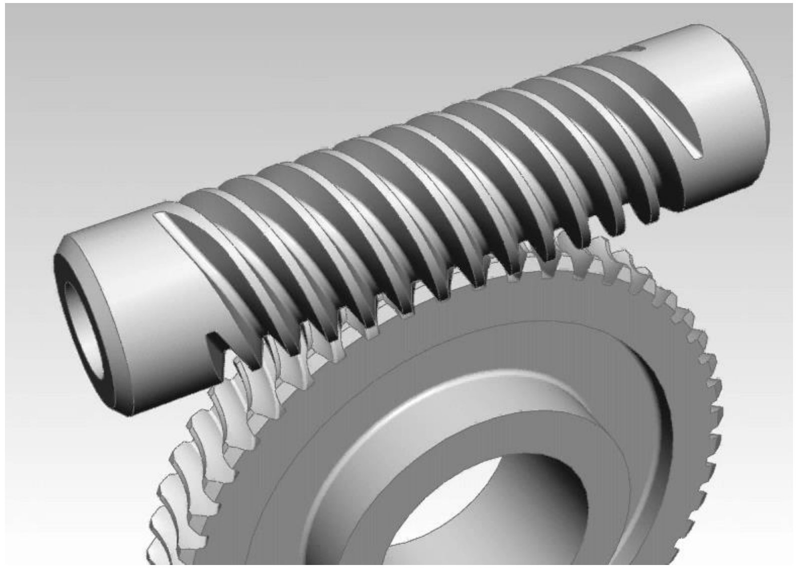
Types of Gears



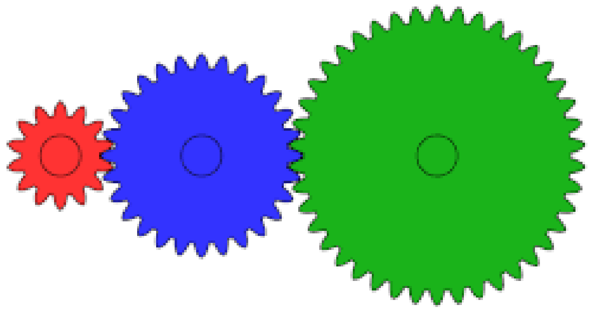






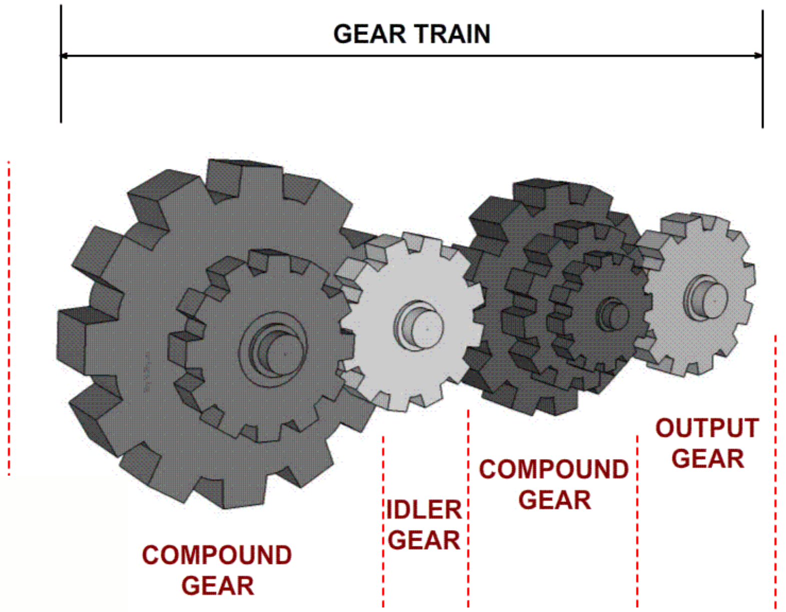


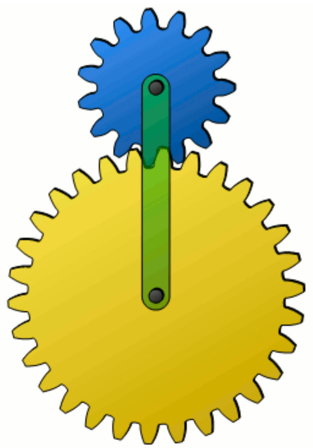
Gear trains

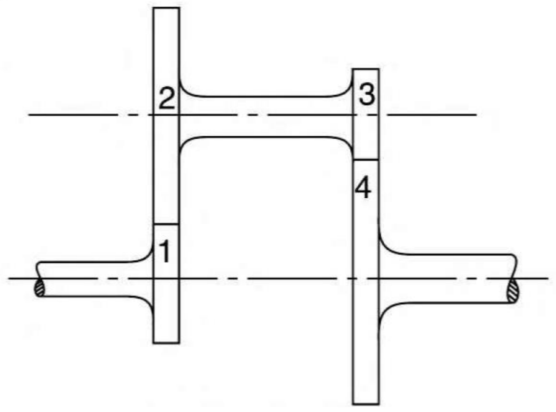


Series of intermeshed gear wheels

- Simple gear train
- Compound gear trains
- Reverted Gear Train
- Epicyclic Gear Train



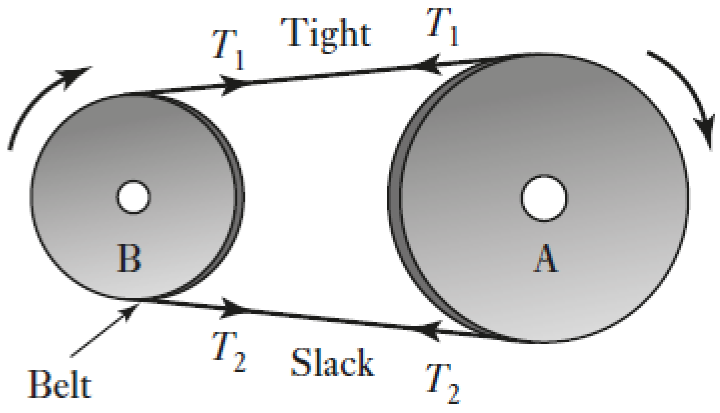


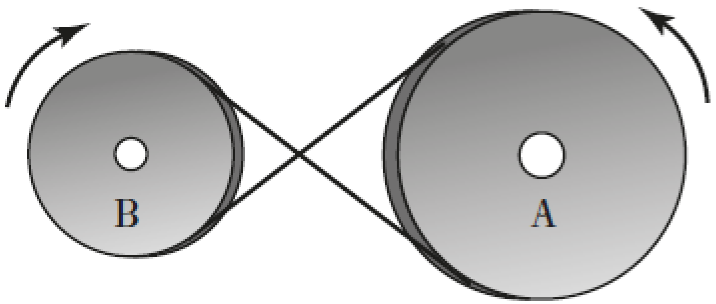


Belt Drive

Pair of rolling cylinders with the motion of one cylinder being transferred to the other by a belt

- Open belt and Crossed belt



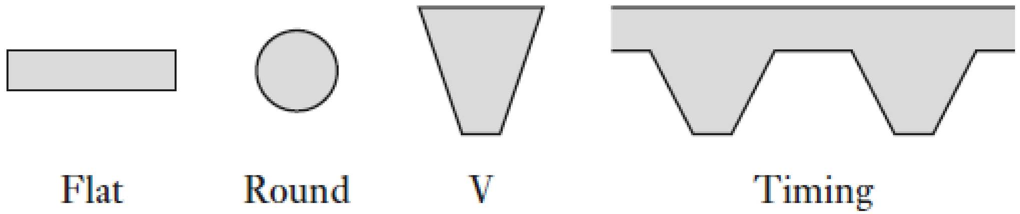


$$\text{torque on A} = (T_1 - T_2)r_A$$

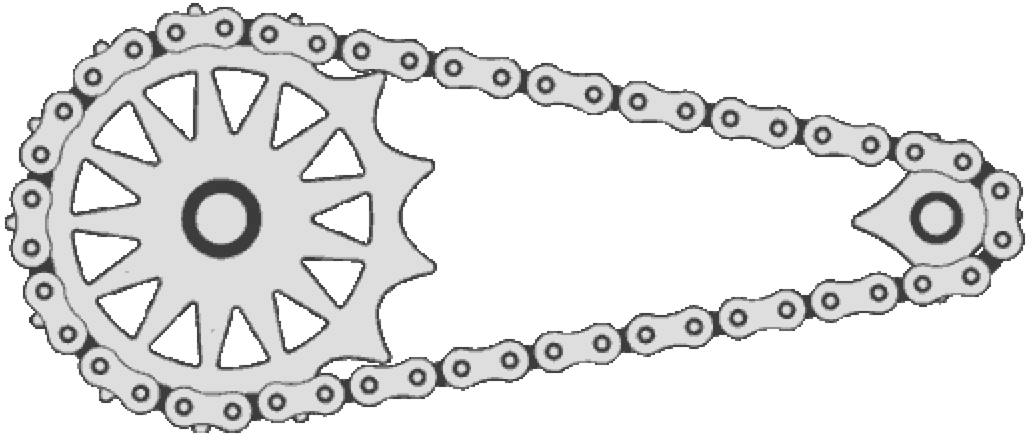
$$\text{torque on B} = (T_1 - T_2)r_B$$

$$\text{power} = (T_1 - T_2)v$$

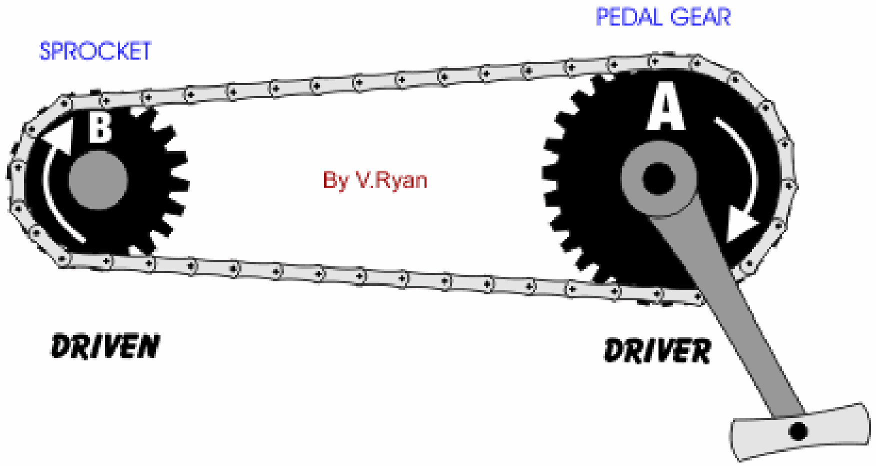
Types of belt.



Chain Drive

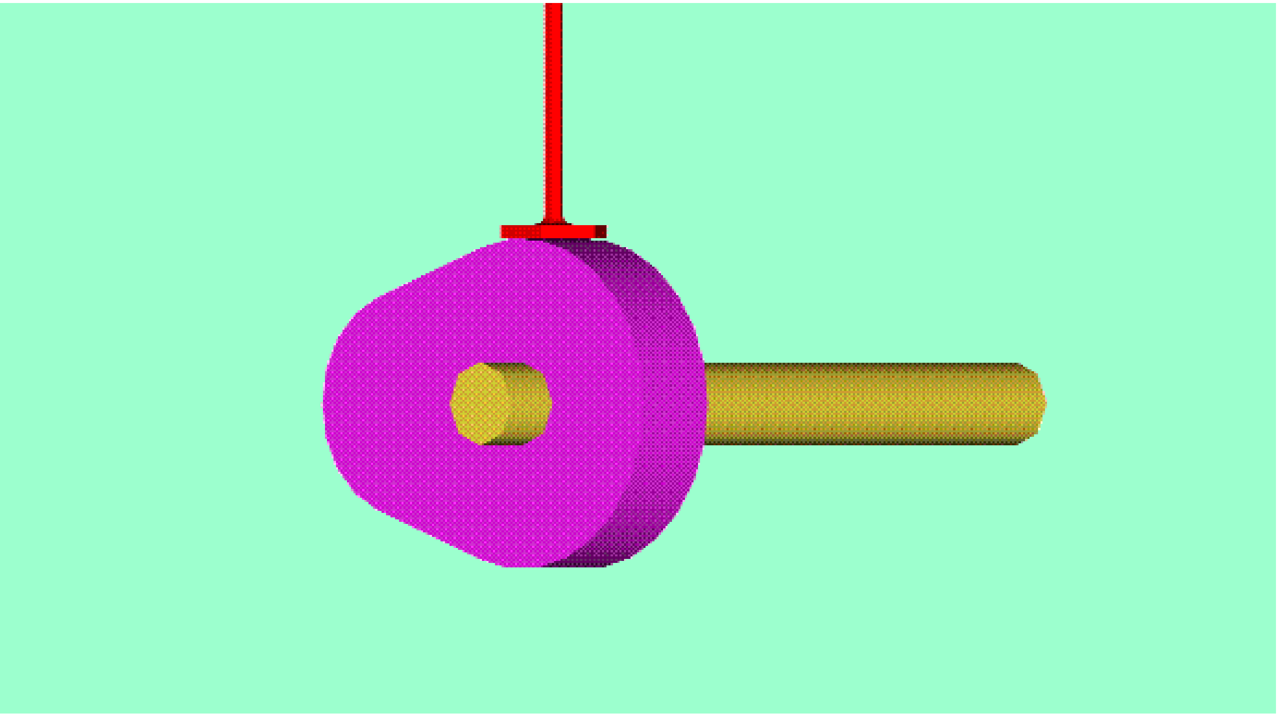


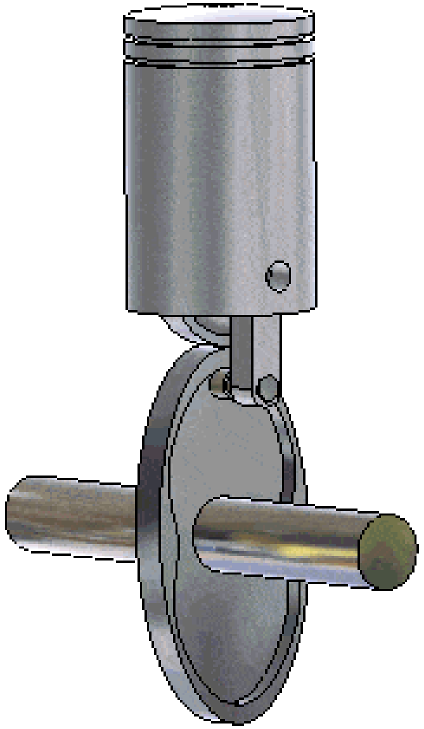
Slip can be prevented by the use of chains which lock into teeth on the rotating cylinders



Cam and follower mechanism

- A cam is ***a mechanical member which impart desired motion to the follower by direct contact***. Cam may be *rotating or reciprocating* while a follower may be *rotating*, *reciprocating and oscillating*. A cam-follower forms higher pair as they have point contact at the junction.





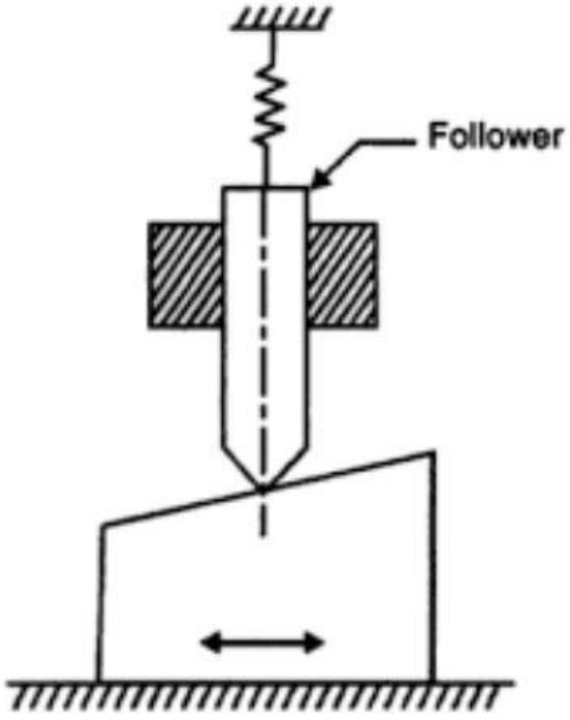
Elements of cam

- Driver member is cam
- The driven member is a follower
- A frame or structure which guides both cam and follower

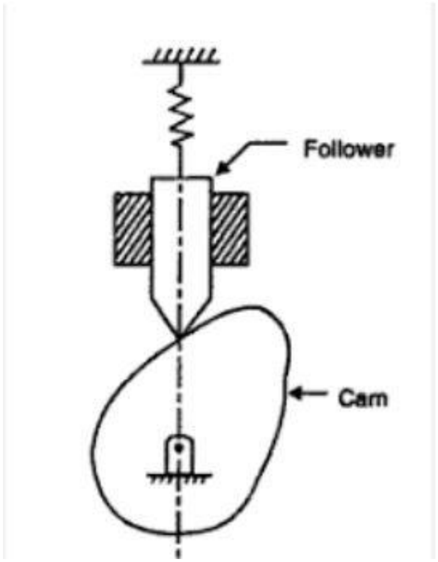
Types of cam

According to shape

- Wedge or flat cam - This cam has transnational motion.
- Radial or disc cam - Here follower moves radically from the center of rotation of the cam.
- Spiral cam - This is also known as face cam as a groove is cut in the form of the spiral.
- Cylindrical cam - A circumferential contour is cut on the cylinder surface which rotates about its axis. It is also known as barrel or drum cam.
- Conjugate cam - This is double disc cam used when we require low wear, noise and better control of follower, high speed and high dynamic load.
- Globoidal cam - It has two types of faces- convex and concave. It is used for moderate speed and angle of oscillation of follower is large.

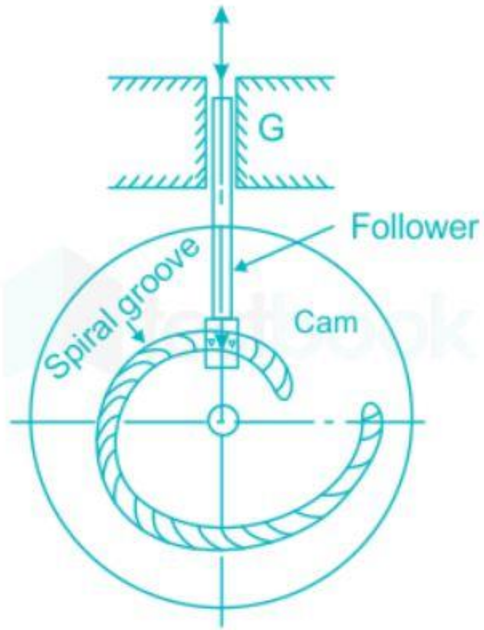


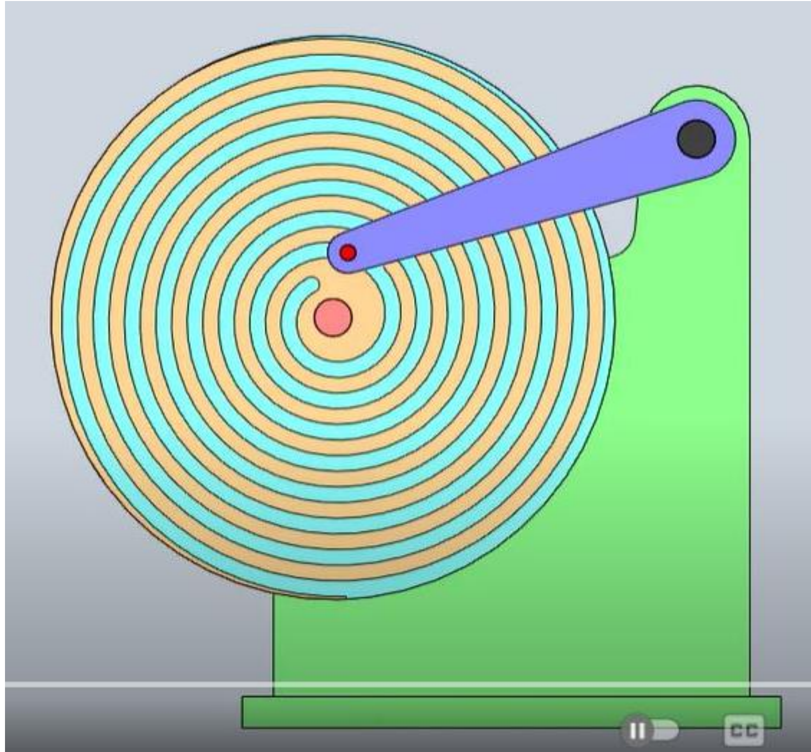
WEDGE CAM

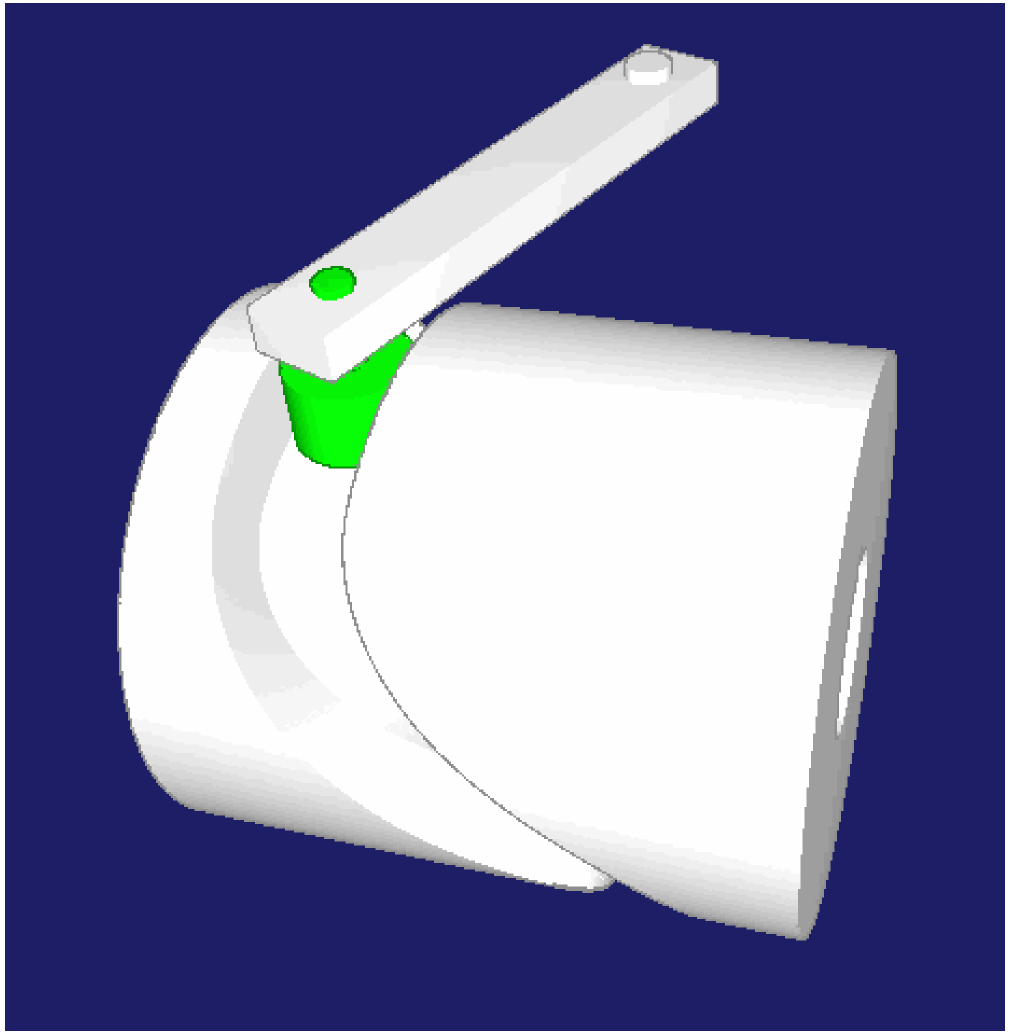


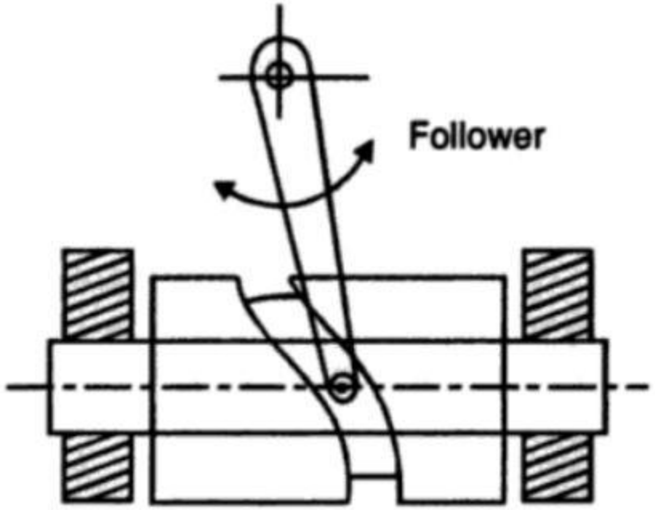
Radial or disc cam

Spiral cam

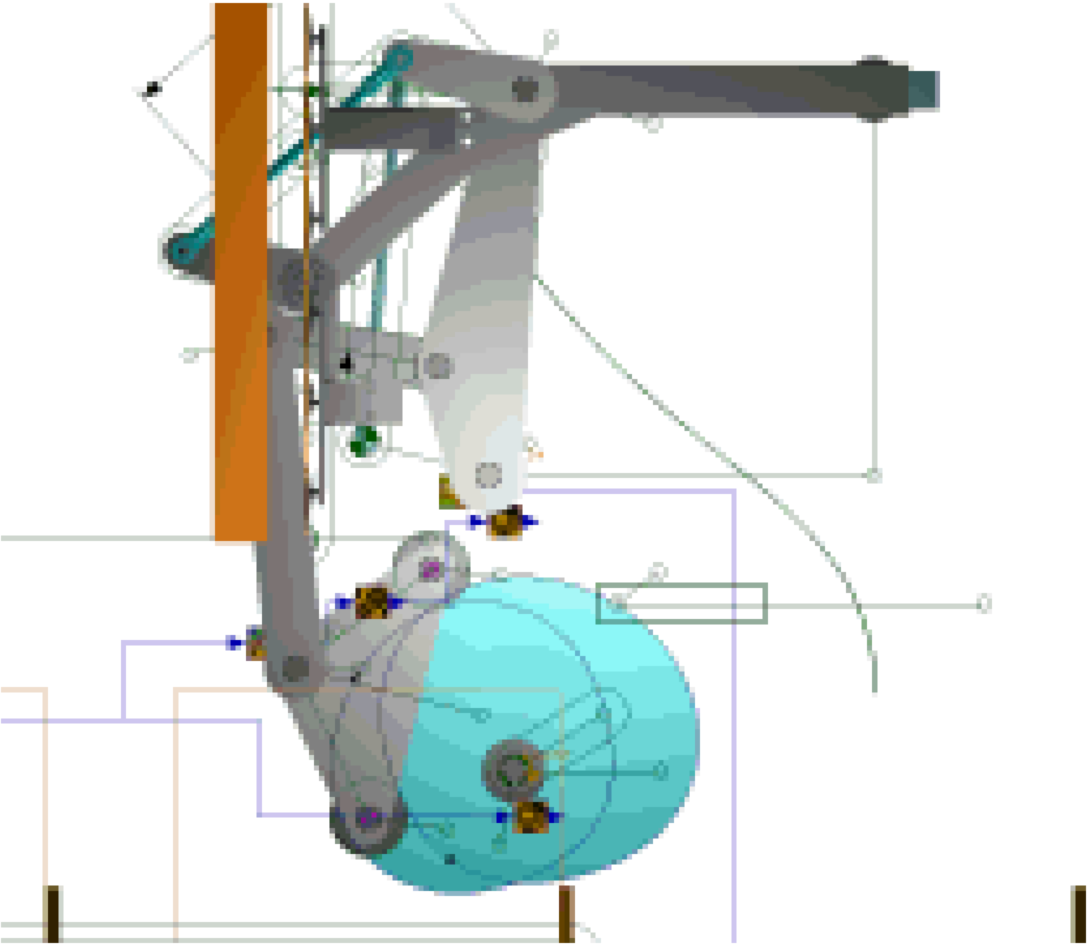


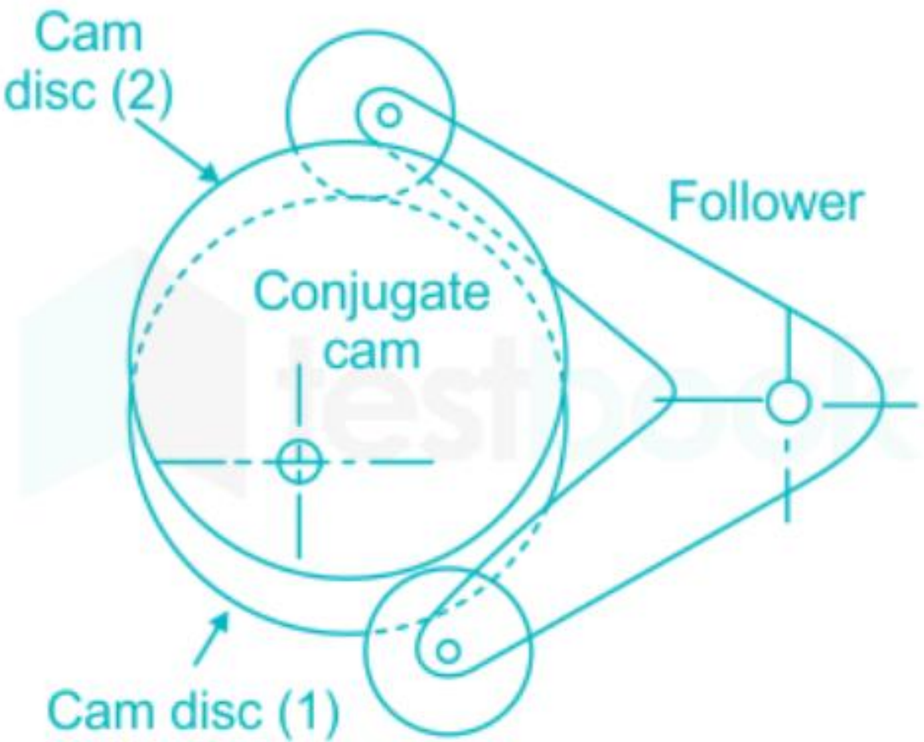






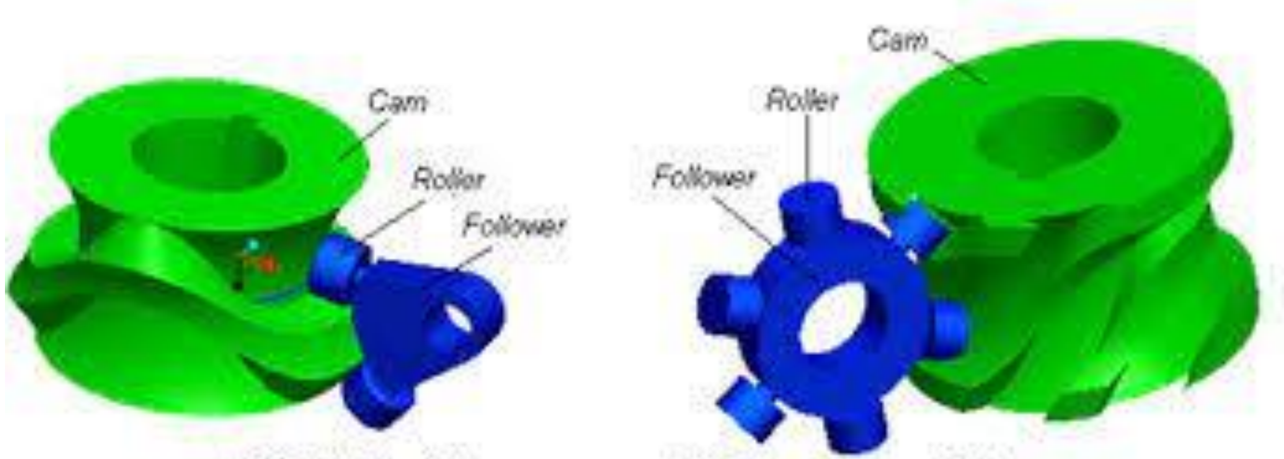
CYLINDRICAL CAM





Conjugate cam

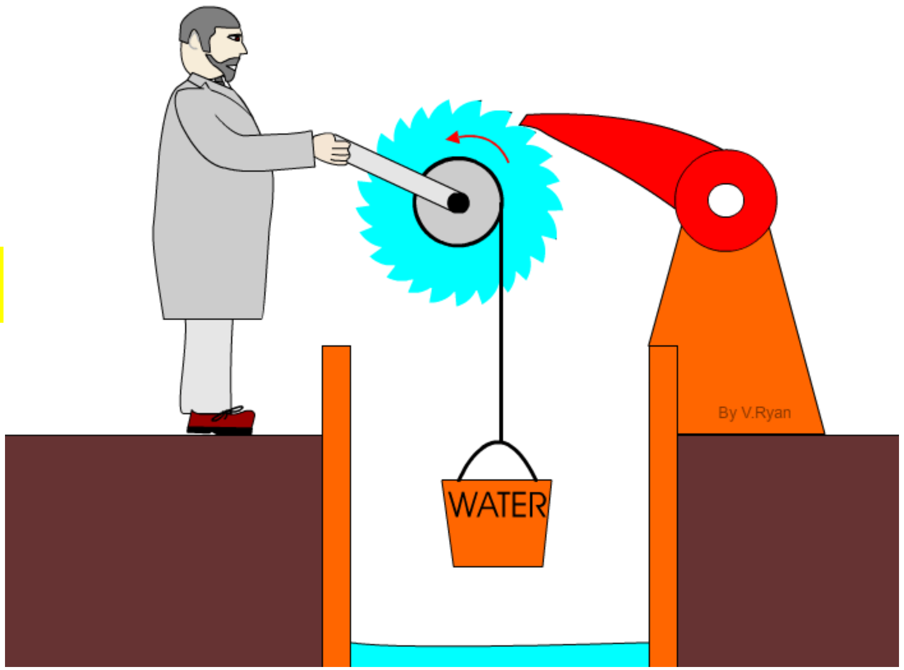
Globoidal cam

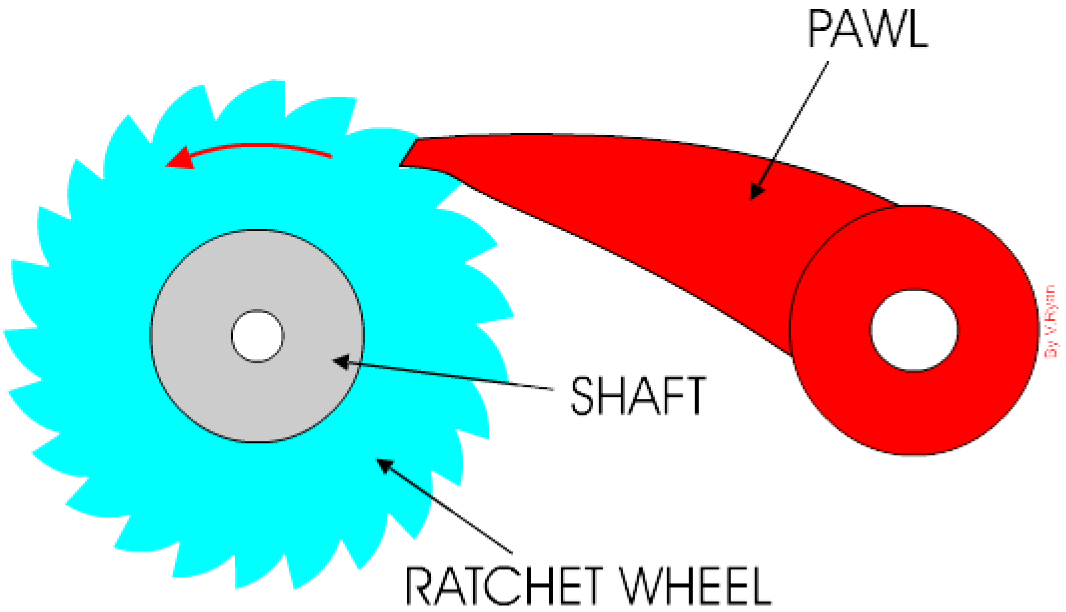


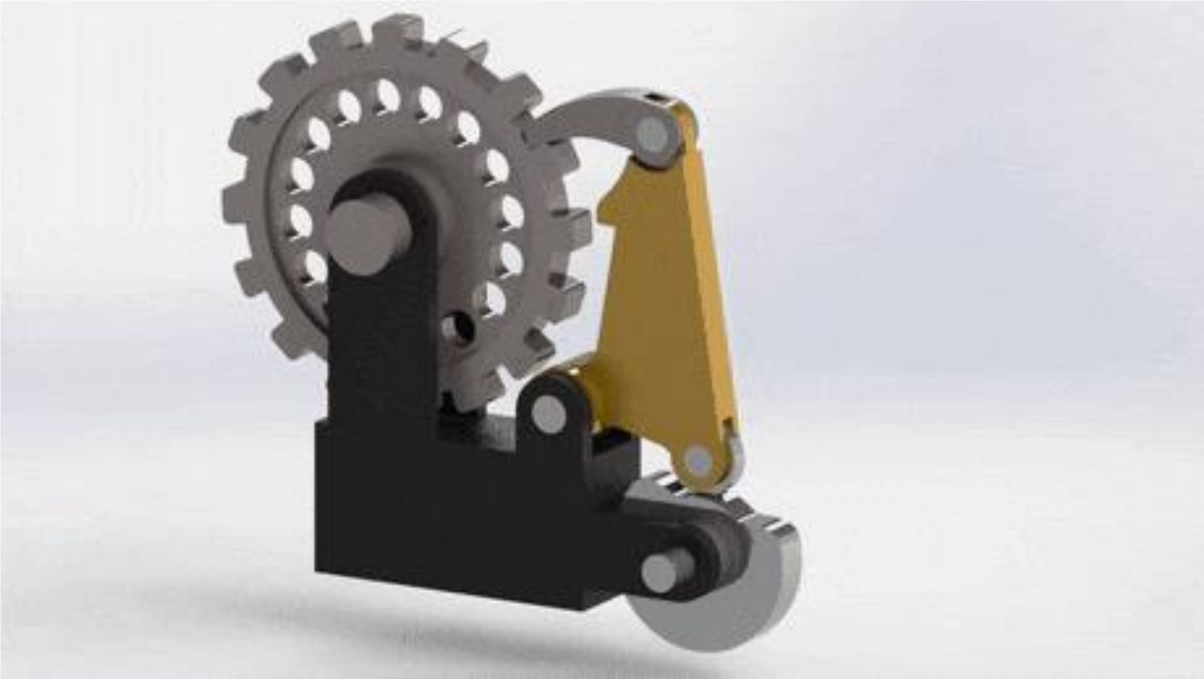
(a) Oscillating follower (b) Indexing turret follower.

Ratchet and pawl

- A ratchet is a device that allows linear or rotary motion in only one direction, while preventing motion in the opposite direction.
- Ratchets consist of a gearwheel and a pivoting spring loaded finger called a pawl that engages the teeth







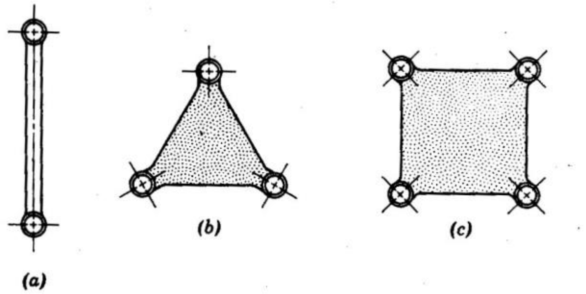
Kinematic chains

Link or element: It is the name given to any body which has motion relative to another.

Binary link: Link which is connected to other links at two points.

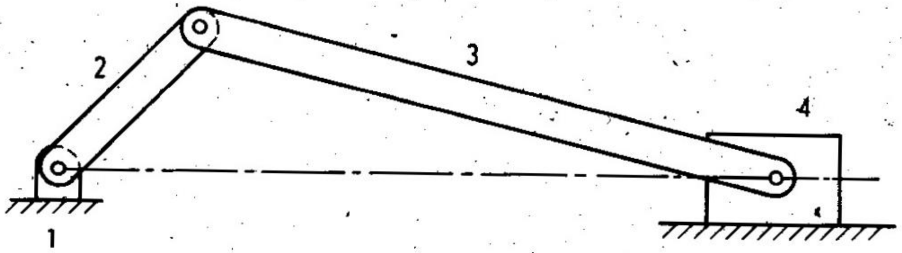
Ternary link: Link which is connected to other links at three points.

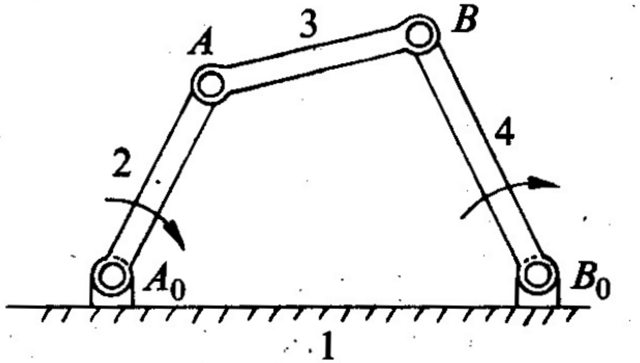
Quaternary link: Link which is connected to other links at four points.



Kinematic chain: A kinematic chain *is a group of links either joined together or arranged in a manner that permits them to move relative to one another*. If the links are connected in such a way that no motion is possible, it results in a locked chain or structure.

Mechanism: A mechanism is a *constrained kinematic chain*. This means that *the motion of any one link in the kinematic chain will give a definite and predictable motion relative to each of the others*. Usually one of the links of the kinematic chain is fixed in a mechanism.

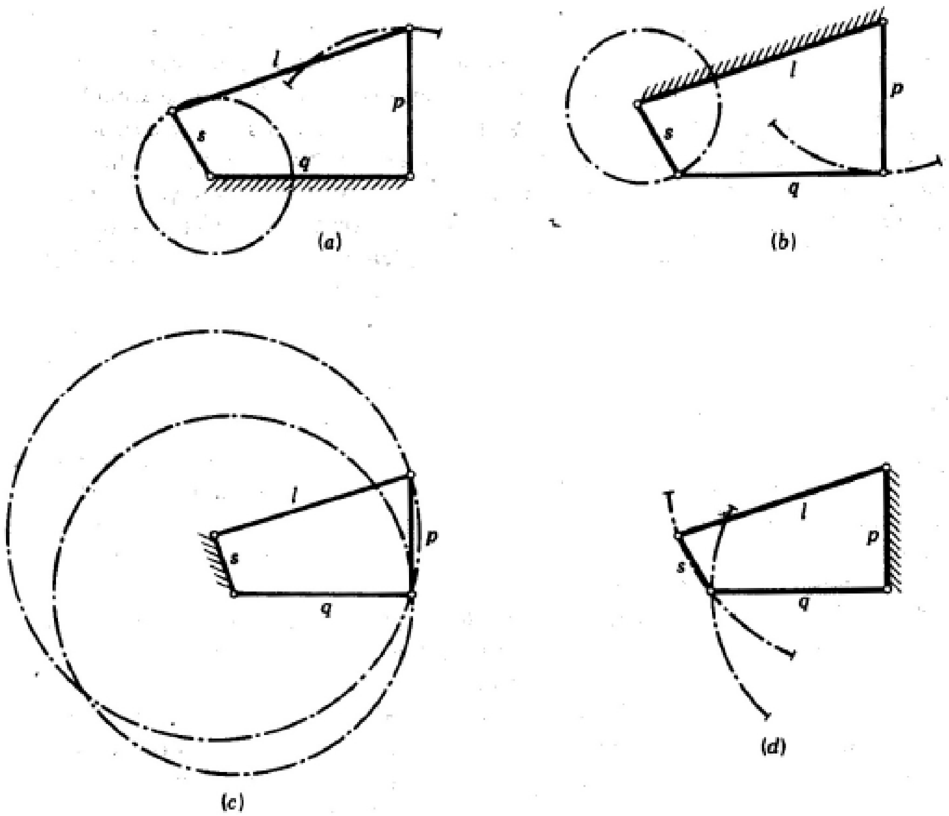




Kinematic chains - Inversions of four bar m/s

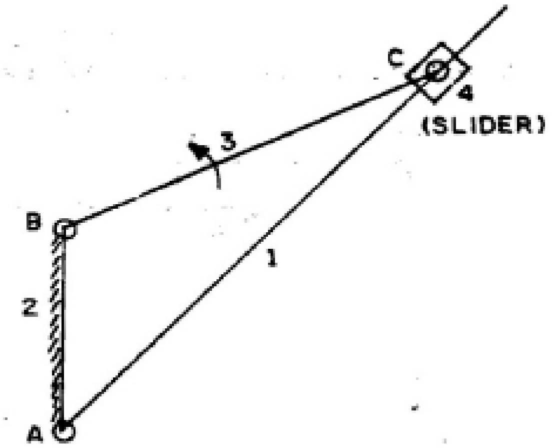
Inversions of mechanism: A mechanism is one in which one of the links of a kinematic chain is fixed. Different mechanisms can be obtained by fixing different links of the same kinematic chain

- Crank-rocker mechanism: link 1 or 3 is fixed, Link 2 (crank) rotates completely and link 4 (rocker) oscillates.
- Drag link mechanism: Here link 2 is fixed and both links 1 and 4 make complete rotation but with different velocities
- Double rocker mechanism: In this mechanism, link 4 is fixed. Link 2 makes complete rotation, whereas links 3 & 4 oscillate

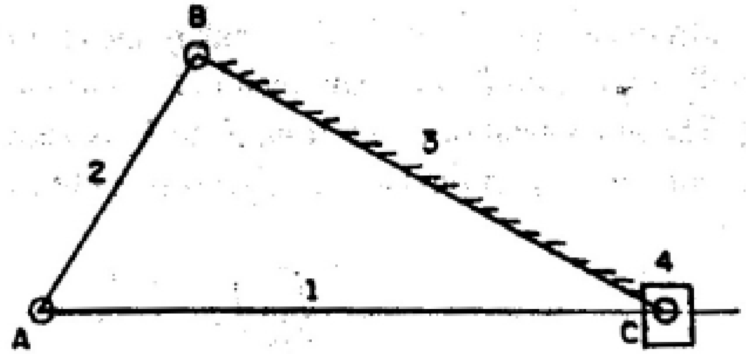


Kinematic chains - Inversions of slider crank m/s

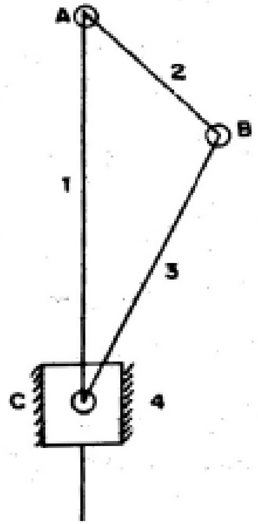
- Inversions of slider crank mechanism is obtained by fixing links 2, 3 and 4



(a) crank fixed



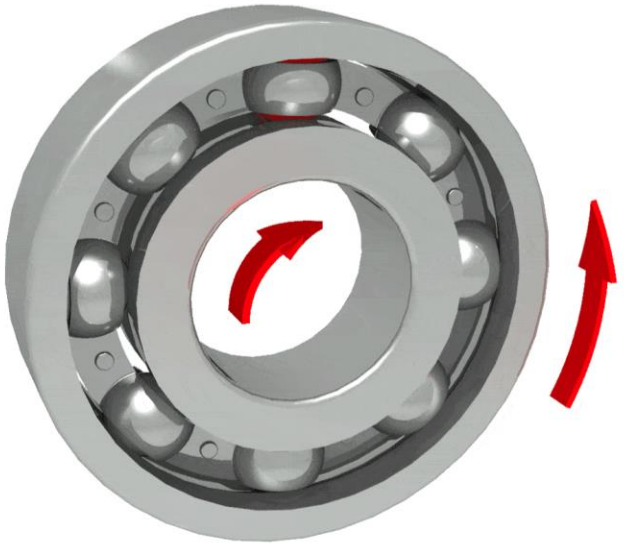
(b) connecting rod fixed

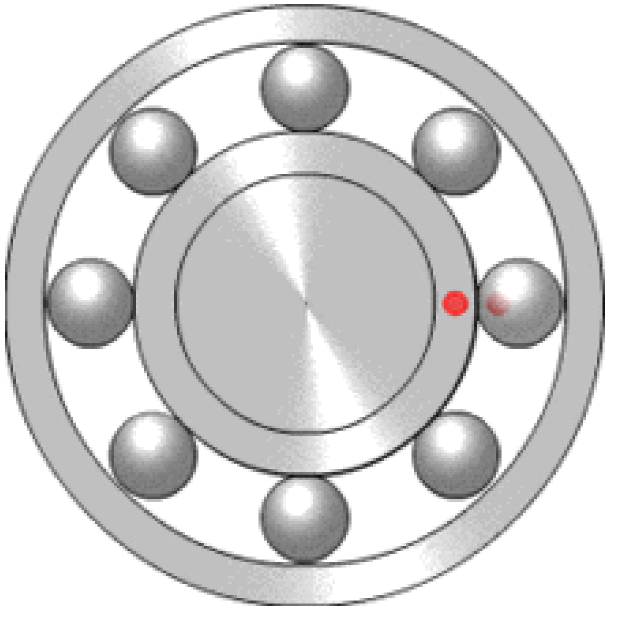


(c) slider fixed

Bearings

- Bearings are mechanical actuators that allow constrained relative motion between objects
  - Bearings allow for smooth rotation with minimal friction.
- Whenever there is *relative motion of one surface in contact with another*, either by *rotating or sliding*, the resulting *frictional forces generate heat which wastes energy and results in wear*
- The function of a bearing is :
  - To guide with minimum friction and maximum accuracy the movement of one part relative to another

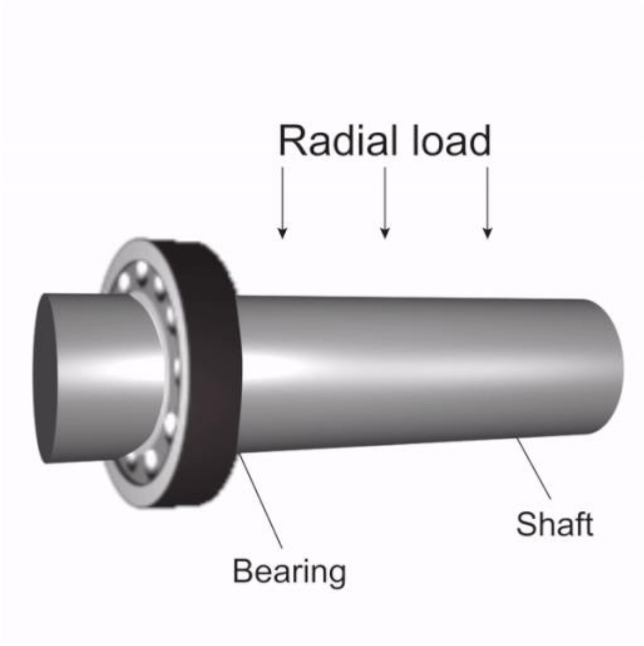


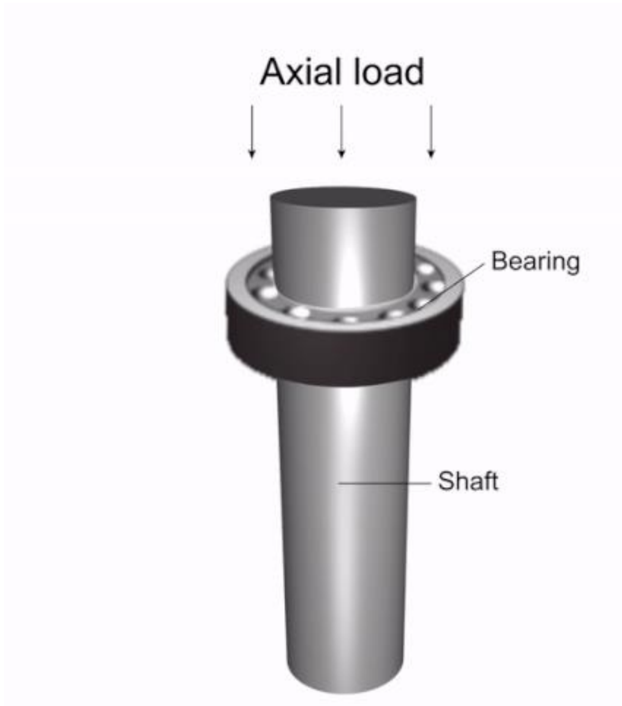


Journal bearing and Thrust bearing

- Journal bearing: Handles radial loads
  - These are forces applied perpendicular to the axis of rotation.
  - Provide suitable support to rotating shafts, i.e. support radial loads

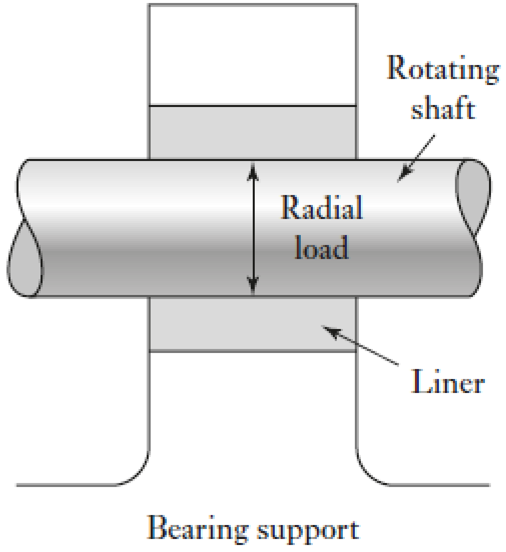
- Thrust bearing: Handles axial loads. These are forces pushing or pulling the shaft along its axis of rotation, parallel to its length.





Journal bearings

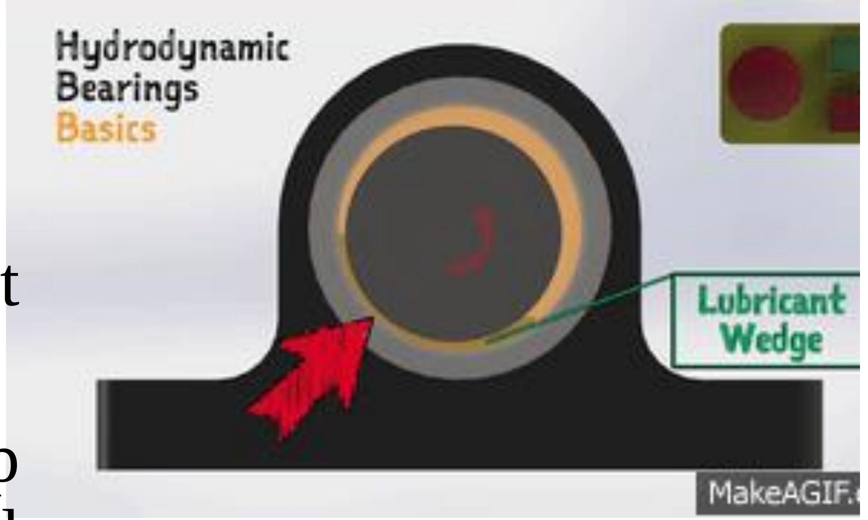
- Journal bearings are used to support rotating shafts which are loaded in a radial direction
- The bearing basically consists of an insert (Liner) of some suitable material which is fitted between the shaft and the support
    - The insert may be a white metal, aluminum alloy, copper alloy, bronze or a polymer
- The bearing may be dry rubbing bearing or lubricated
    - Plastics such as nylon generally used without lubrication
    - Lubricated bearing (fluid film bearing) classified as
        - *Hydrodynamic*
        - *Hydrostatic*



Solid Journal bearing

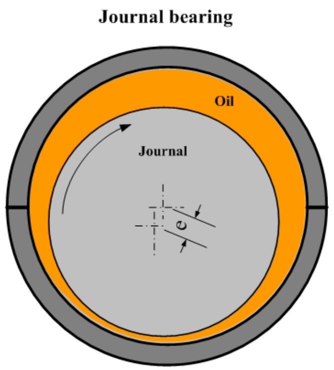
Split Journal bearing

Fluid film bearing



Hydrodynamic

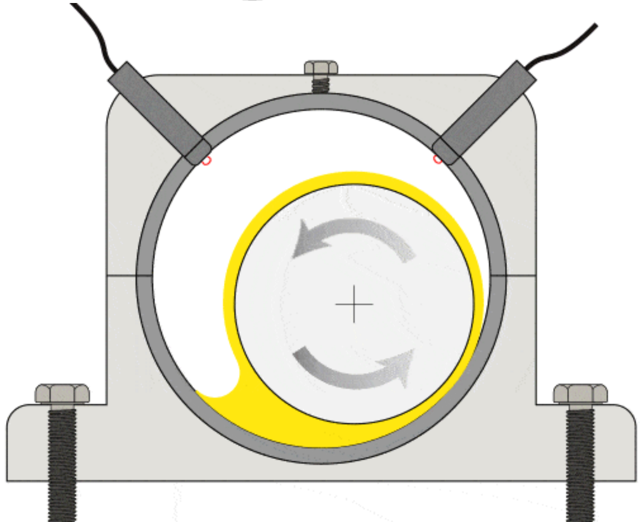
- The shaft is rotating continuously in oil such a way that it rides on oil and is not supported by metal
- As the shaft rotates, it creates a wedge-shaped gap between itself and the bearing. The lubricant gets dragged into this wedge, increasing pressure and creating a lifting force that separates the shaft from the bearing



Hydrostatic

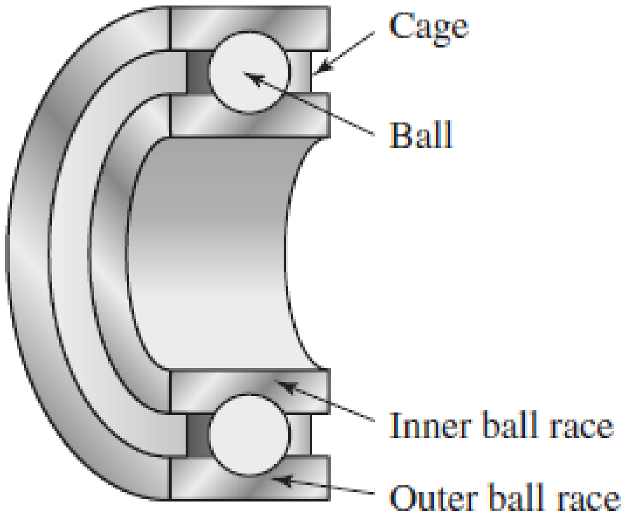
A problem with hydrodynamic lubrication is that the shaft only rides on oil when it is rotating and when at rest there is metal-to-metal contact

To avoid excessive wear at start-up and when there is only a low load, oil is pumped into the load-bearing area at a high-enough pressure to lift the shaft off the metal when at rest



Ball and roller bearings

- The main load is transferred from the rotating shaft to its support by rolling contact rather than sliding contact.
- A rolling element bearing consists of four main elements:
  - Inner race
  - Outer race
  - Rolling element (balls or rollers)
  - Cage to keep the rolling elements apart



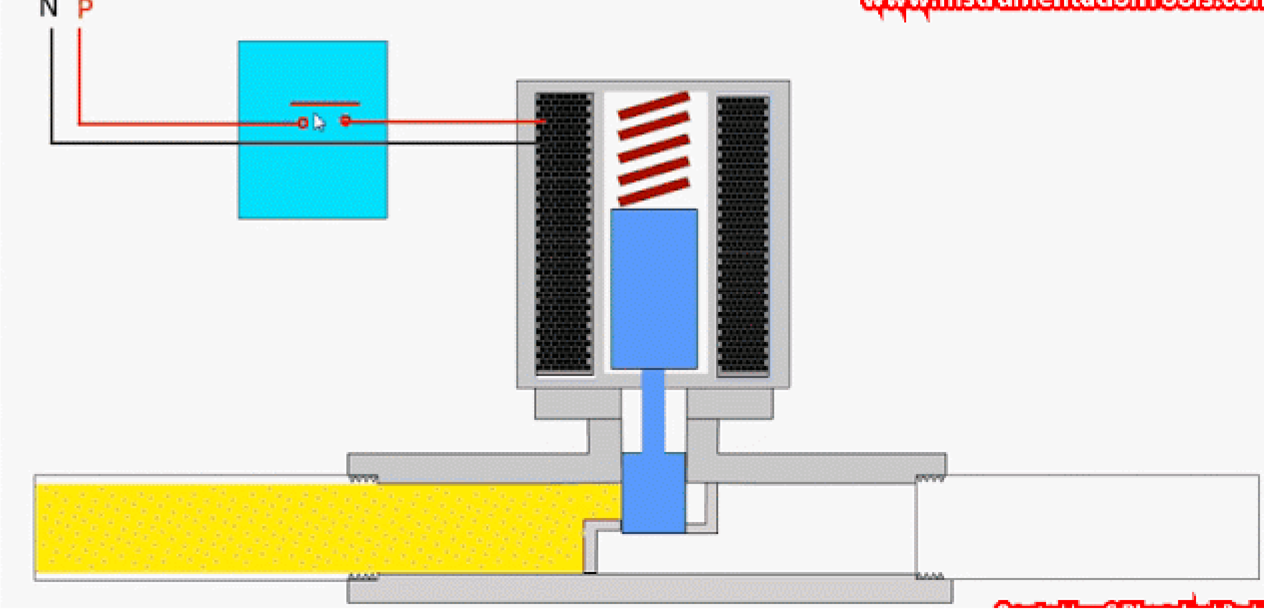
More about bearing, check the link

- https://www.youtube.com/watch?v=8q25EUszBSI&ab_channel=TheEngineersPost
- https://www.youtube.com/watch?v=Xb59xt_bkok&ab_channel=TechnicalEngineeringSchool

Electrical actuation systems

1. Switching devices such as mechanical switches e.g. relays, and solid-state
   - Relays are electrical components (*electromechanical devices with moving parts*) that control the flow of current in a circuit.
   - Solid-State switches are Electronic devices with no moving parts. Use semiconductors like transistors or thyristors to switch current
2. Solenoid-type devices
3. Drive systems, such as d.c. motors and a.c. motors, where a current through a motor is used to produce rotation.

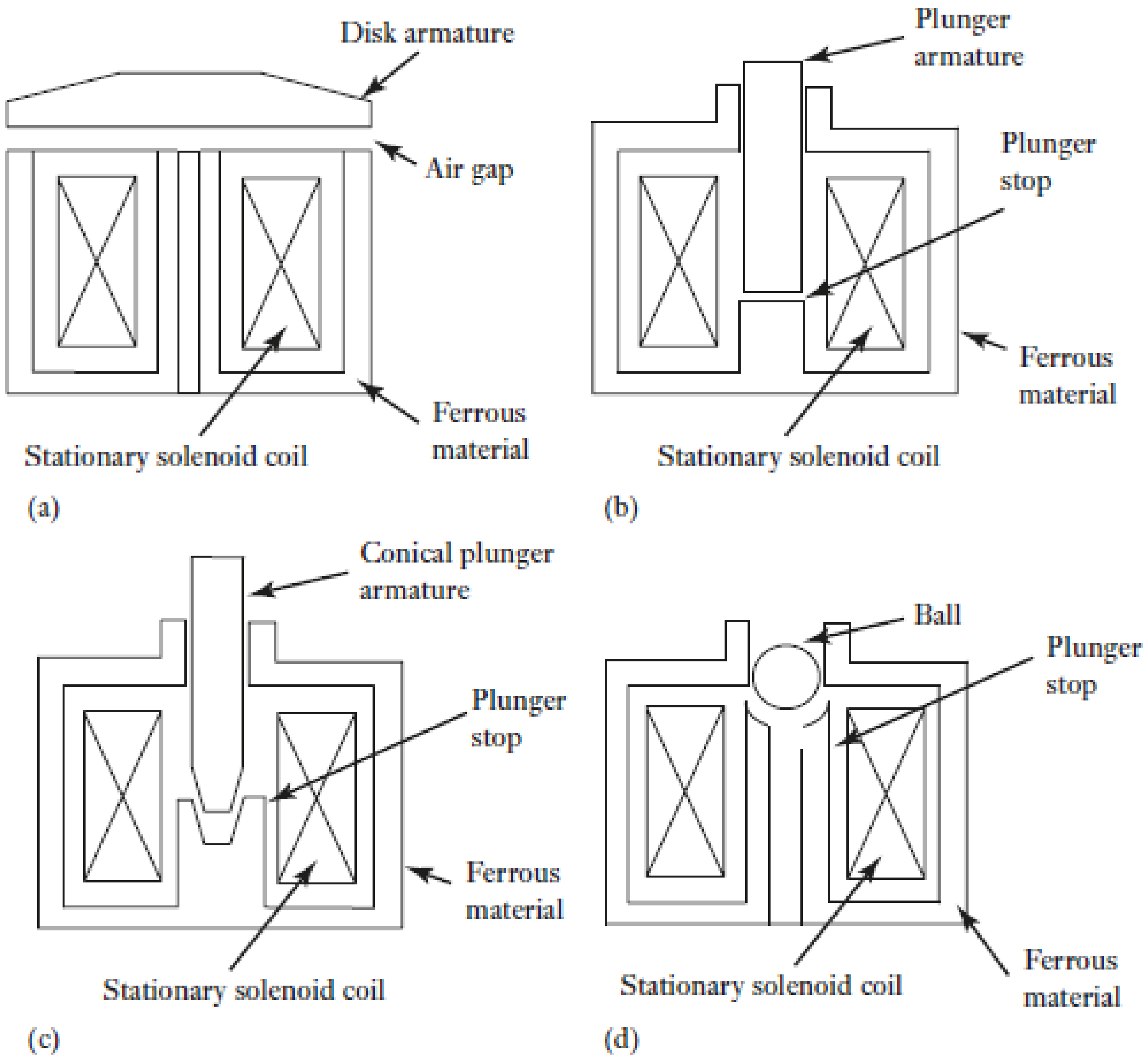
Solenoids



- Solenoids consist of a coil of electrical wire with an armature which is attracted to the coil when a current passes through it and produces a magnetic field.
- The movement of the armature contracts a return spring which then allows the armature to return to its original position when the current ceases

Examples of linear solenoids with <u>different forms of armature</u>

- Disk-
  - Useful where small distances of travel and fast action are required
- Plunger
- Conical plunger –
  - Used for long-stroke applications
  - A typical application being for an automotive doorlock mechanism
- Ball forms of armature
  - Fluid control applications



Latching solenoid actuator

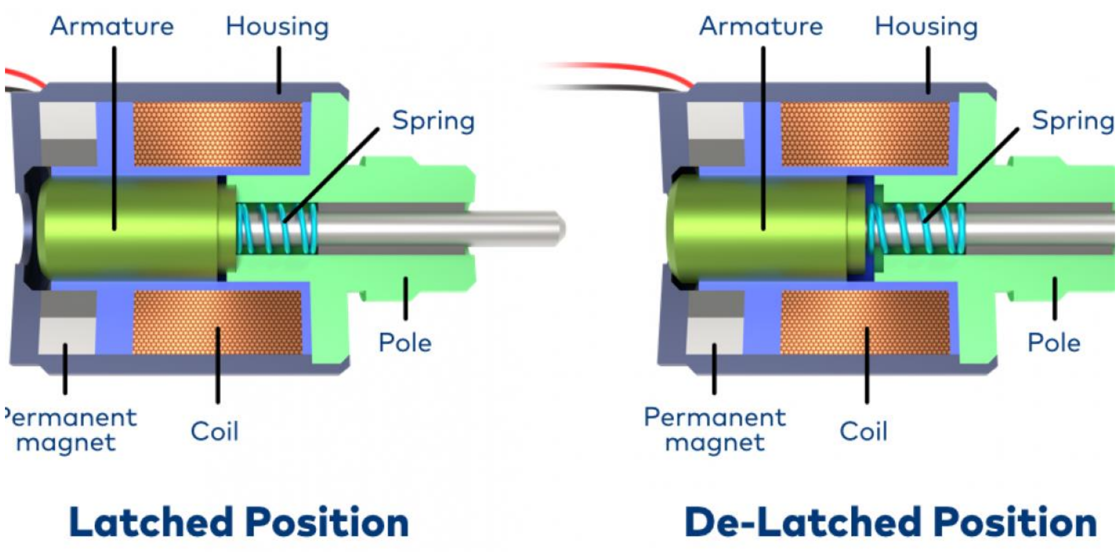
- Solenoid actuators can be made to be latching (retain their actuated position) when the solenoid current is switched off
- A permanent magnet is added so that when there is no current through the solenoid it is not strong enough to pull the armature against its retaining spring into the closed position.

❖ Once the armature has moved full travel and is in contact with the pole, it will remain in this position without any further electrical power input.

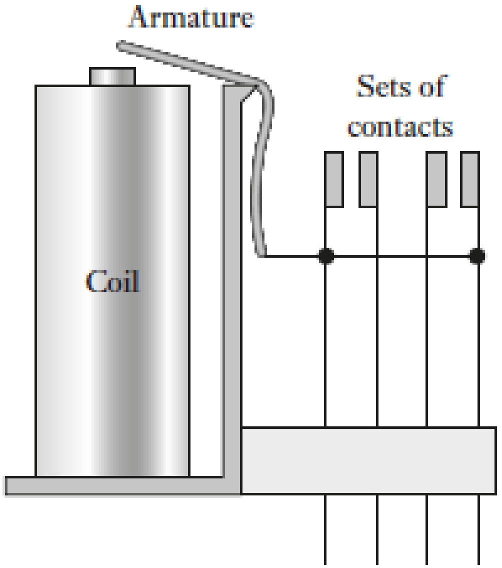
❖ The armature is held in this position by the permanent magnet.

❖ To release the solenoid from this hold position, the "holding" magnet's attraction has to be canceled by sending a current back through the coil field in the opposite direction.

❖ latching solenoids can <u>maintain the extended or retracted position without consuming continuous power</u>



Relay Switch



- An electromagnetic device.
- It uses a low-power control circuit to trigger an electromagnet.
- This electromagnet then physically opens or closes contacts in a higher-power circuit, acting like a switch.
- when there is a current through the solenoid of the relay, a magnetic field is produced which attracts the iron armature, moves the push rod, and so closes the normally open(NO) switch contacts and opens the normally closed (NC) switch contacts.
- Relays are often used in control systems.
- It controls a high-power circuit by using a low-power control circuit.
- When a current is applied to the coil of the relay, it creates a magnetic field that switches the contacts on or off, connecting or disconnecting the high-power circuit.

Solid state switches – Fundamentals of electronics

A PN junction is a fundamental building block in electronics. It's formed by joining two specially treated pieces of semiconductor material together, creating a special boundary.

Semiconductors: These are materials that can conduct electricity under certain conditions.

- They are neither good conductors like metals nor perfect insulators.
- Doping, adding impurities, can further change their electrical properties.

P and N type semiconductors: Doping creates two types of semiconductors,

- P-type with an abundance of holes (positive charge carriers)
- N-type with an excess of electrons (negative charge carriers)

- The Junction: When a P-type and N-type region are joined, a depletion zone forms where charges from each side diffuse and cancel each other. This creates a potential barrier.

The key property of a PN junction is its ability to rectify current:

- Biasing: ***Applying an external voltage to the junction affects current flow***
- Forward Bias: If the positive voltage is connected to the P-type side and negative to N-type, it reduces the depletion zone allowing current to flow easily.
- Reverse Bias: When the positive voltage is connected to the N-type and negative to P-type, it widens the depletion zone significantly blocking current flow.

Solid state switches

- Solid state switches are electronic components that perform the function of switching
- Unlike mechanical switches that have moving parts, solid state switches are faster, more reliable, and can handle higher currents.
- Four common types of solid state switches are
  - ❖Diodes
  - ❖Thyristors and Triacs
  - ❖Bipolar transistors
  - ❖power MOSFETs

Solid state switches

Diodes:

The simplest type of solid state switch. They allow current to flow in one direction only

Thyristors and Triacs:

These are three-terminal devices that act like switches for AC and DC current respectively.

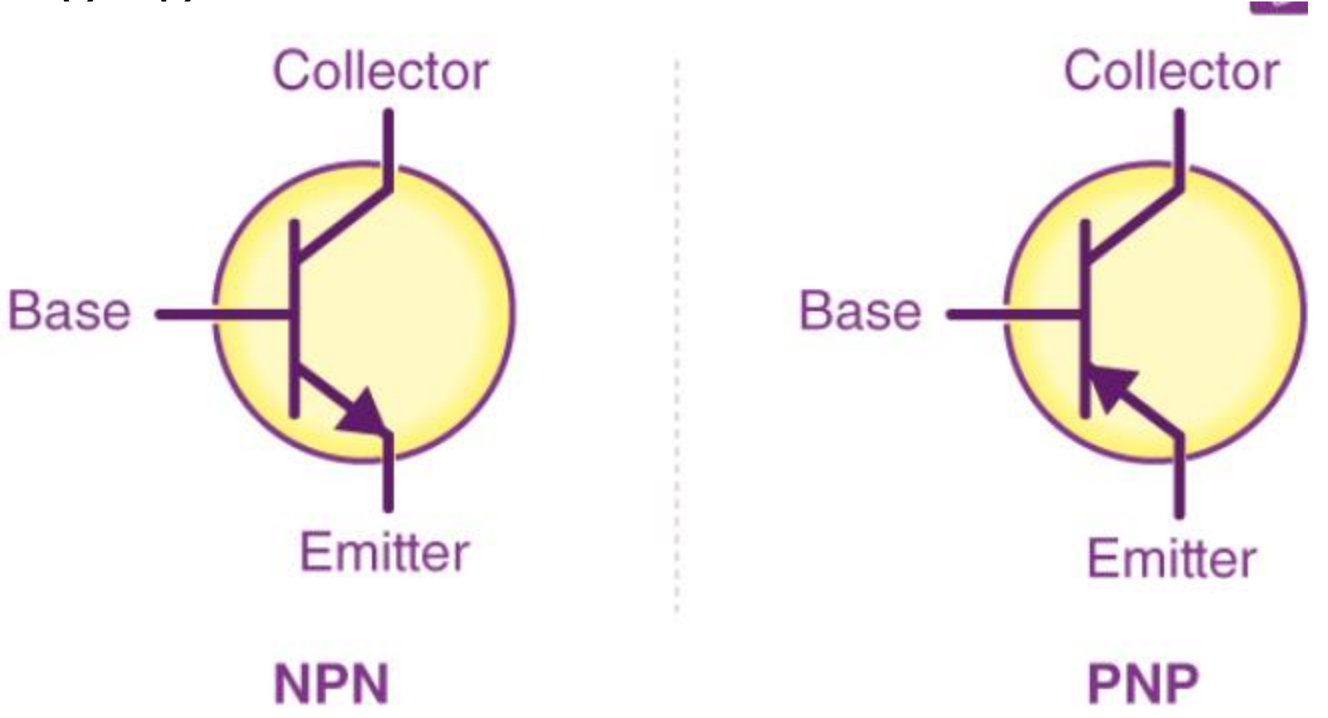
- Thyristors can only be turned on, not off
- Triacs can be turned on and off

They are useful for controlling high power AC loads.

Solid state switches

Bipolar Junction Transistors (BJTs):

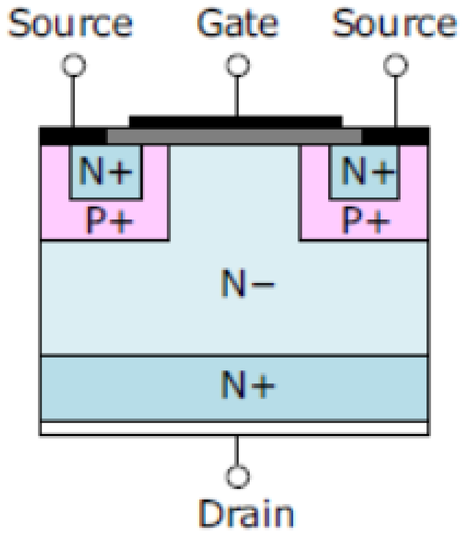
- BJTs are fundamental building blocks of electronic circuits.
- They *use a small current at the base terminal to control a much larger current flowing between the collector and emitter terminals*.
- BJTs are used for amplifying or switching signals in electronic devices.



Solid state switches

Power MOSFETs:

- Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs) are voltage-controlled devices.
- By *applying a voltage to the gate terminal, the resistance between the drain and source terminals is changed, allowing current to flow or blocking it*.
- Power MOSFETs are widely used in electronic circuits for power control due to their fast switching speed and high efficiency.



DC and AC motors

DC motors

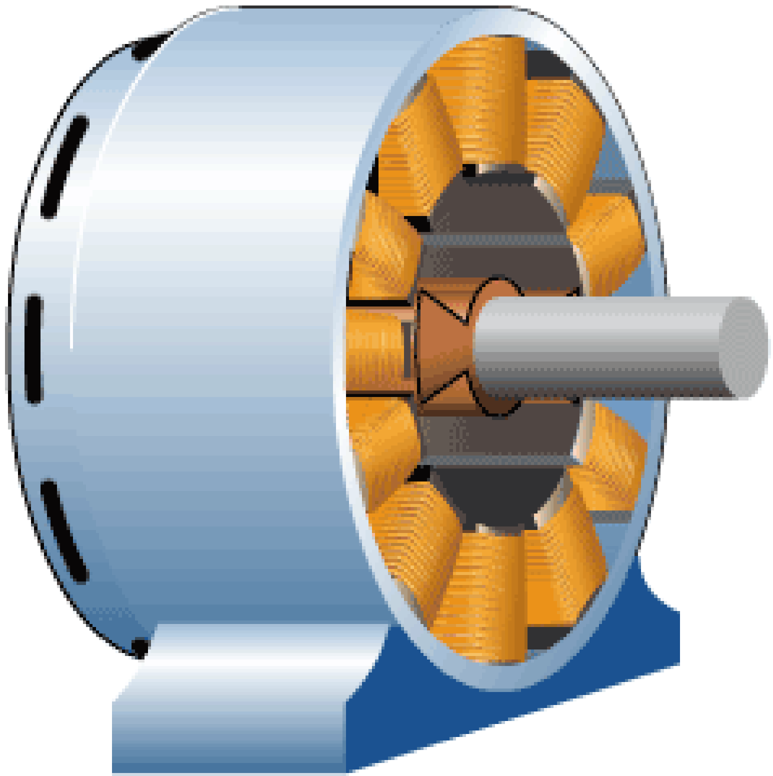
- ❖Brush-type dc motor
- ❖Brush-type dc motors with field coils
  - o *Series wound motor*
  - o *Shunt-wound motor*
- ❖Brushless permanent magnet dc motors

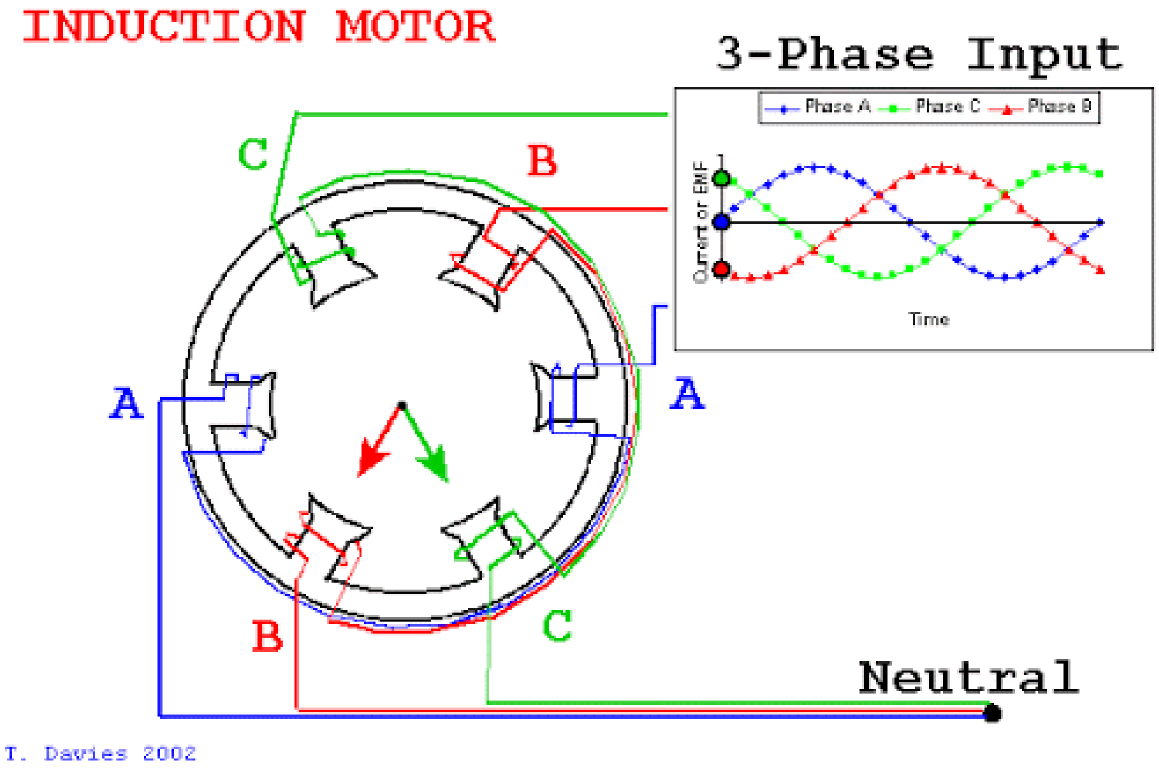
AC Motors

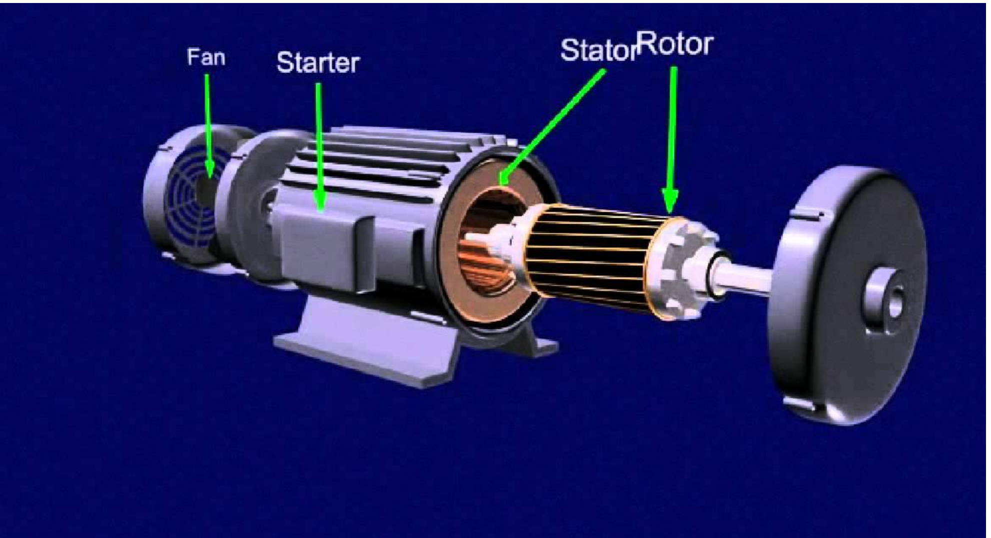
- ❖Single-phase induction motor
- ❖Three-phase induction motor
- ❖Three-phase synchronous motor

Stepper motors

AC motor

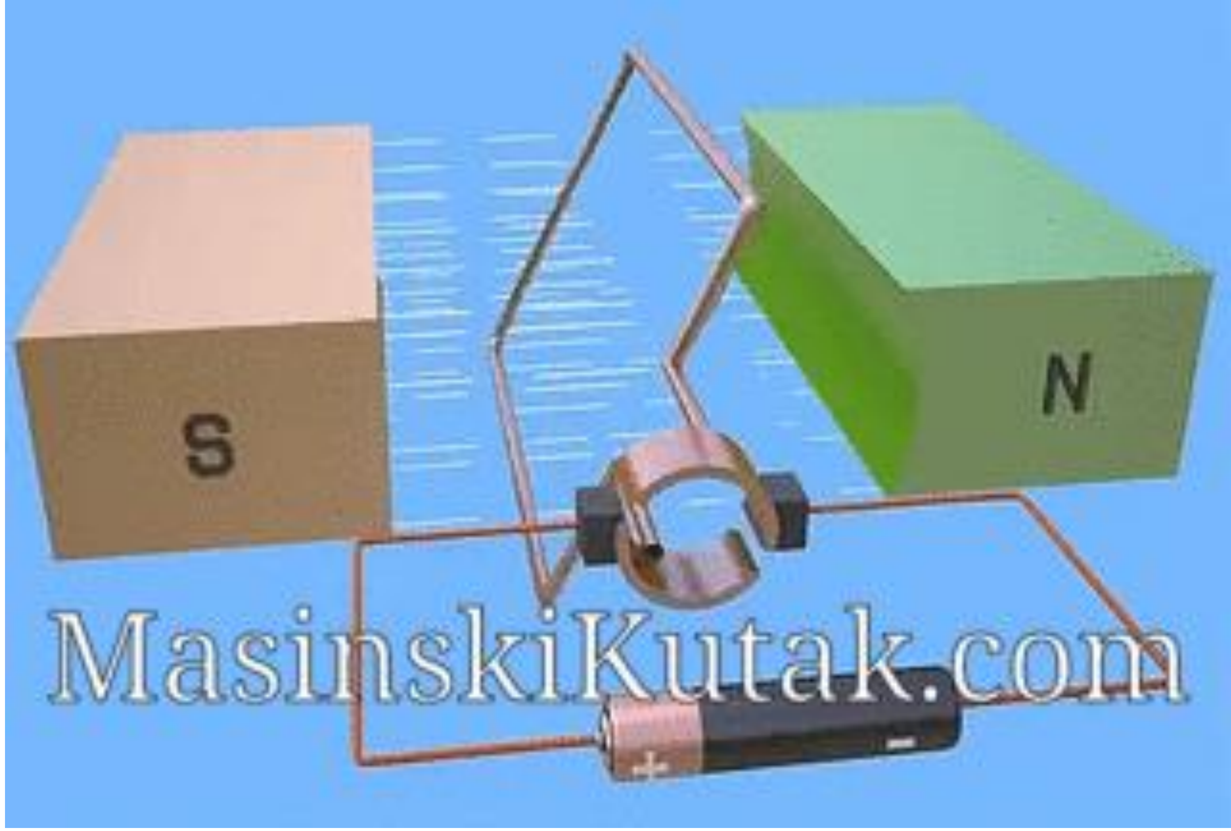






Power Source	AC current	DC current
Complexity	Simpler design	More complex design
Cost	Less expensive	More expensive
Speed Control	More difficult	Easier
Common Applications	Household appliances, industrial machinery	Battery-powered devices, electric vehicles

DC Motor



The main parts of a DC motor are:

- **Stator:** The stationary part of the motor that holds the permanent magnets or electromagnets which create the magnetic field.
- **Armature:** The rotating part of the motor, typically made up of coils of wire wrapped around a soft iron core.
- **Commutator:** A split ring-like device made of copper segments that is attached to the rotating armature. It makes electrical contact with the brushes.
- **Brushes:** Sliding contacts that conduct current between the commutator and the external circuit